

Review

Review of Family-Level Short-Term Load Forecasting and Its Application in Household Energy Management System

Ping Ma ^{1,†}, Shuhui Cui ^{1,†}, Mingshuai Chen ², Shengzhe Zhou ³ and Kai Wang ^{1,*} 

¹ School of Electrical Engineering, Weihai Innovation Research Institute, Qingdao University, Qingdao 266000, China; qdumaping@163.com (P.M.); cuishuhui321@163.com (S.C.)

² Rizhao Power Supply Company, State Grid Shandong Electric Power Company, Rizhao 276826, China; chenmingshuaizsz@163.com

³ Department of Information Engineering, Shandong Water Conservancy Vocational College, Rizhao 276826, China; zhoushengzheyx@126.com

* Correspondence: wangkai@qdu.edu.cn or wkwj888@163.com; Tel.: +86-15863060145; Fax: +86-532-85951980

† These authors contributed equally to this work.

Abstract: With the rapid development of smart grids and distributed energy sources, the home energy management system (HEMS) is becoming a hot topic of research as a hub for connecting customers and utilities for energy visualization. Accurate forecasting of future short-term residential electricity demand for each major appliance is a key part of the energy management system. This paper aims to explore the current research status of household-level short-term load forecasting, summarize the advantages and disadvantages of various forecasting methods, and provide research ideas for short-term household load forecasting and household energy management. Firstly, the paper analyzes the latest research results and research trends in deep learning load forecasting methods in terms of network models, feature extraction, and adaptive learning; secondly, it points out the importance of combining probabilistic forecasting methods that take into account load uncertainty with deep learning techniques; and further explores the implications and methods for device-level as well as ultra-short-term load forecasting. In addition, the paper also analyzes the importance of short-term household load forecasting for the scheduling of electricity consumption in household energy management systems. Finally, the paper points out the problems in the current research and proposes suggestions for future development of short-term household load forecasting.

Keywords: home energy management systems; household-level load forecasting; short-term load; deep learning neural networks; probabilistic forecasting



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1. Introduction

With the development of the economy and living standards, household-based electrical energy consumption accounts for an increasing proportion of total energy consumption, and how to use electrical energy more efficiently is the key to reducing residential energy consumption. In addition, with the development of energy saving and emission reduction policies and distributed energy sources, energy management with the home energy management system (HEMS) as the main energy saving countermeasure has gradually become the focus of residential energy optimization research [1]. Accurate forecasting of future residential electricity demand in the short term is an intelligent decision-making tool for energy management systems. It is therefore important to study methods for predicting household-level and even device-level loads with high forecasting accuracy.

Load forecasting methods can generally be classified according to the forecasting time scale: long-term load forecasting (LTLF), medium-term load forecasting (MTLF), short-term load forecasting (STLF), and very short-term load forecasting (VSTLF). Among them, short-term load forecasting can provide effective energy system planning, which is important for energy management [2,3]. Traditional short-term load forecasting has focused

on the system level, substation level, busbar level, feeder level, and building level. Unlike system-level electrical loads, individual household loads lack a clear consistent pattern conducive to forecast accuracy. In general, the diversity of aggregation levels smooths out the daily load distribution, which makes substation loads relatively more predictable, while individual household electricity consumption is more dependent on underlying human behavior. Within individual households, the daily habits and lifestyles of residents may have a more direct impact on the short-term load distribution. The randomness and uncertainty caused by human behavior or weather changes make it difficult to accurately implement more established system-level load forecasting methods at the household level.

With the widespread installation of smart meters, household electricity consumption data are available and household-level load forecasting is possible. Accurate short-term load forecasting (STLF) can properly complement the HEMS and support utilities to implement more rational demand response strategies to reduce operational costs [4]. In recent years, in order to further improve the accuracy of short-term household load forecasting, there have been many related studies at home and abroad, proposing various forecasting methods including probabilistic load forecasting and various deep learning algorithms. The aim of this paper is to review the research papers in this field, compare the accuracy and advantages and disadvantages of various forecasting methods, and give effective suggestions for the research of household-level load forecasting methods.

The rest of the paper is organized as follows:

1. Section 2 introduces the application of deep learning algorithms such as RNN, LSTM, and CNN in household load forecasting, and analyzes the improvement of the algorithms in terms of fusion of models, feature analysis and extraction, and adaptive learning of networks.
2. Section 3 explores the advantages of probabilistic forecasting methods and proposes a research direction to improve traditional probabilistic forecasting methods with advanced deep learning algorithms.
3. Section 4 analyzes the development trends and challenges of ultra-short-term device-level load forecasting.
4. Section 5 presents the principles of the HEMS and the key role played by home-level load forecasting in the HEMS.

2. Deep Learning-Based Load Prediction Methods

There are many research methods on load forecasting, from classical time series analysis to more recent machine learning methods, including the Autoregressive Integrated Moving Average Model (ARIMA) [5], Support Vector Regression (SVR) [6], Random Forest (RF) [7], and Gradient Boosting Regression (GBR) [8]. Another important branch of load forecasting is the use of artificial neural network (ANN) models [9] for forecasting. In recent years, with the increase in relevant load data and massively parallel computing capabilities, load prediction using big data has become a hot research topic. Against the backdrop of massive data and significant growth in parallel computing performance, deep learning (DL) is approaching or even surpassing humans in areas such as speech recognition and image classification and is gradually becoming the model of choice for many researchers to improve load prediction methods.

2.1. Neural Network Prediction Models

In recent years, deep learning has become one of the most active techniques in many areas of research. In contrast to shallow learning, deep learning typically refers to stacking multiple layers of neural networks and relying on stochastic optimization to perform machine learning tasks. Different numbers of layers can provide different levels of abstraction to improve learning ability and task performance. In particular, long short-term memory (LSTM) recurrent neural networks (RNNs), proposed by Hochreiter and Schmidhuber [10], have received a great deal of attention in the field of sequence learning.

2.1.1. LSTM Model

Often, deep learning models encounter the problem of gradient explosion (i.e., learning scattering) or gradient disappearance (i.e., learning stopping). This problem is solved by LSTM networks that introduce memory cells and computational gates. LSTM is a type of RNN that has been used in the past for time series analysis and load prediction problems.

A load profile can be thought of as a time series in which there is a high degree of correlation and dependence between successive samples. Long short-term memory (LSTM) has the structure of a recurrent neural network with the characteristics of sequence learning [2]. The LSTM structure has a memory unit that allows it to remember the significant states of the previous steps. In addition, LSTM has an oblivion gate to remove unimportant features and reset the memory unit, as shown in Figure 1. LSTM can also track customer consumption behavior by learning long-term dependencies and correlations between successive samples of sequential data.

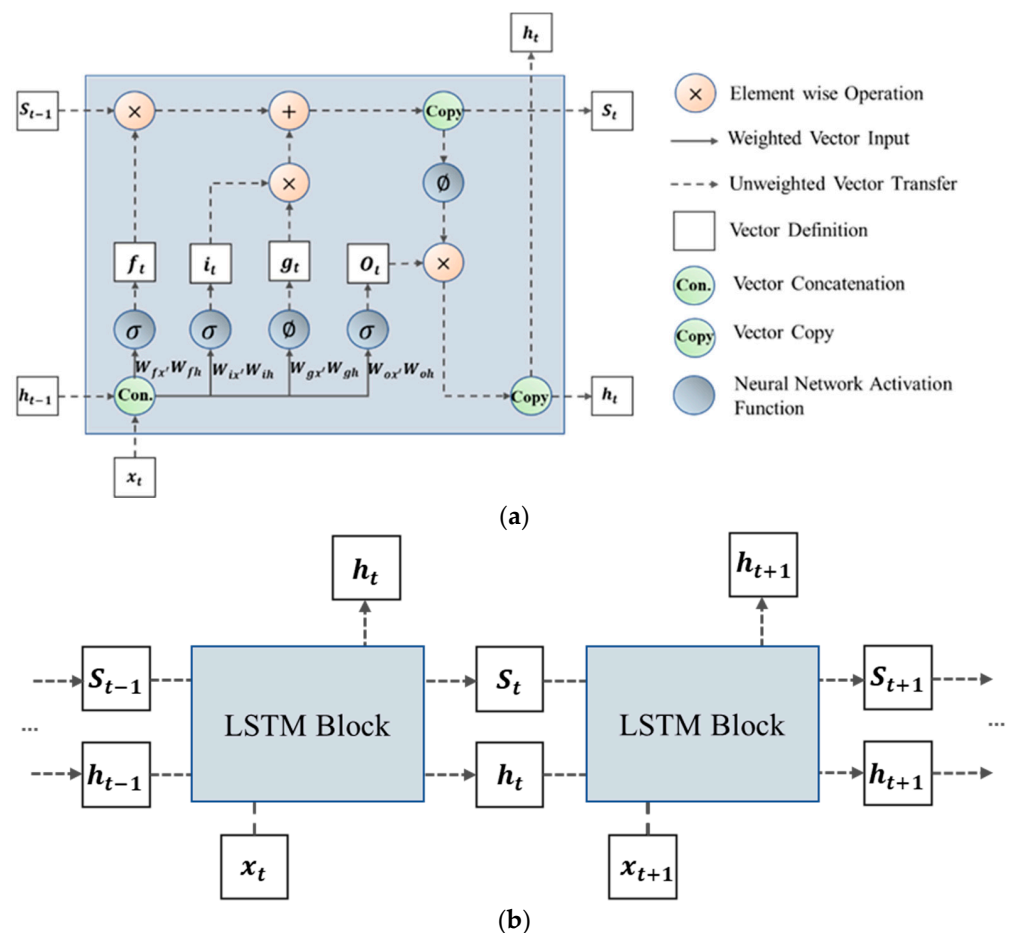


Figure 1. The structure of an LSTM: (a) The structure of an LSTM block. (b) The unrolled LSTM sequential architecture.

It is generally assumed that the lifestyles of the inhabitants, however inconsistent, will be reflected in repeated patterns of energy consumption. Kong et al. [11] proposed a load prediction framework based on an LSTM cyclic neural network (Figure 2). The results show that the prediction accuracy of LSTM is better than several other advanced machine learning algorithms. In the later work, people try to improve the LSTM algorithm in order to obtain higher prediction accuracy. In view of the uncertainty and non-linear characteristics of household load data, the EMD method can transform non-stationary non-linear data into stationary linear data. Liu et al. [12] proposed a combination algorithm of empirical mode decomposition (EMD) and stack long short-term memory (SLSTM) that has been applied to

household short-term load prediction. This SLSTM can train better parameters and improve the accuracy of prediction. SVR has the characteristic of linearly reflecting input–output relationships. Combined with this characteristic, Moradzadeh et al. [13] proposed an improved hybrid method of support vector regression—long-term and short-term memory (SCR–LSTM) for short-term load prediction of a microgrid. The SVR–LSTM model with the highest correlation coefficient ($R = 0.9901$) and the lowest error value can provide better results than the traditional SVR and LSTM. However, the data set used to verify the method is not load data for single households and does not account for weather and human uncertainties. Nevertheless, hybrid approaches such as these are widely proposed.

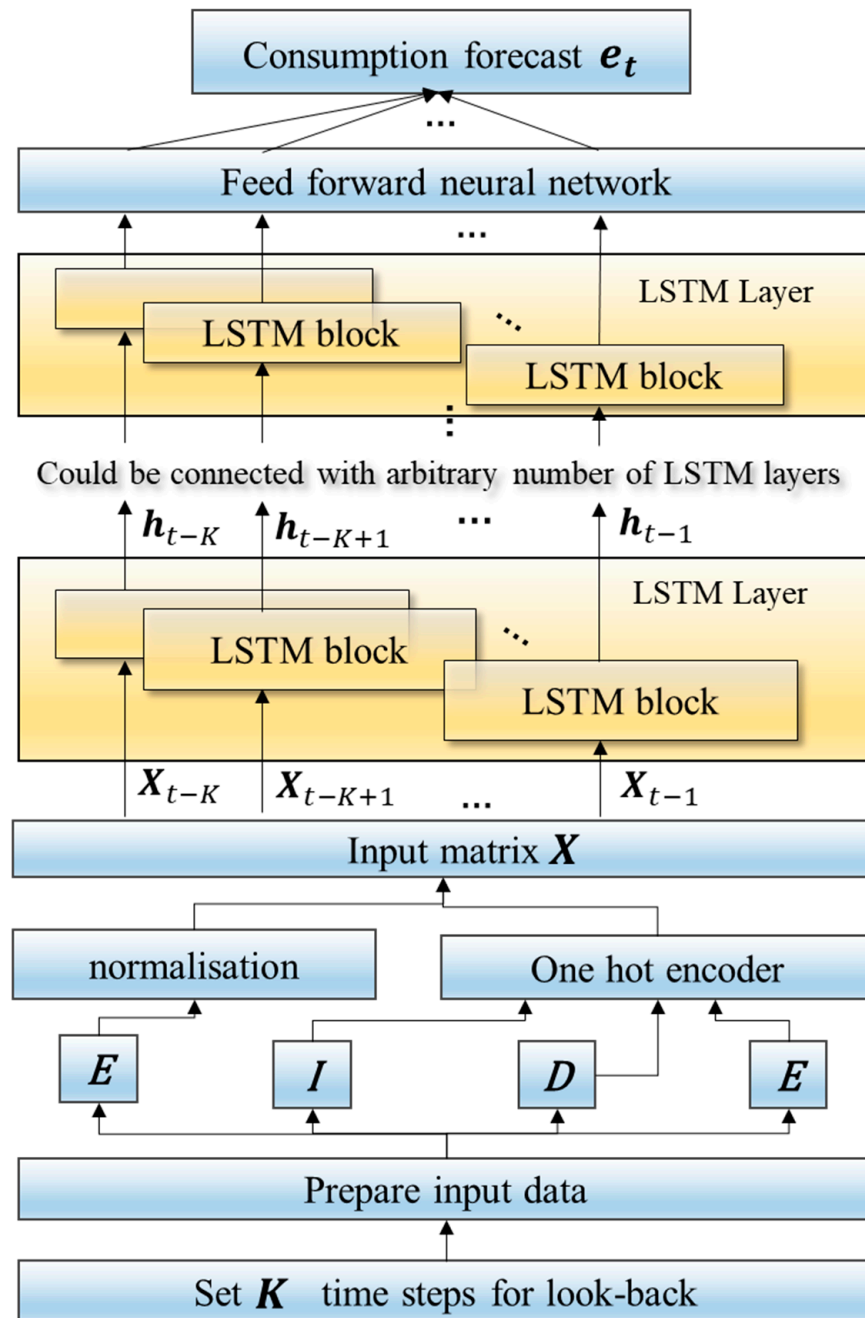


Figure 2. The LSTM-based forecasting framework.

2.1.2. CNN Model

CNN is a type of deep neural network which is most commonly used to analyze visual images. CNN uses the method of local connection and weight sharing to process the original data at a higher level and more abstract, which can effectively and automatically extract the internal features in the data. The internal neural network layer is mainly composed of a convolutional layer, pooling layer, and full connection layer, which reduces the number of weights and network complexity. CNN automatically extracts feature vectors from data, effectively reducing the complexity of feature extraction and data reconstruction, improving the quality of data features, and obtaining effective information through a convolution layer and pooling, greatly reducing the training time of the model.

Since LSTM utilizes the time locality of load time series by analogy with the spatial locality in image processing, CNN seems to be able to obtain better accuracy when we lack historical data, the size of the energy entity is small, and the prediction range is short (forecast one day in advance). Based on the combination of convolutional neural networks (CNNs) and data enhancement technology, Acharya et al. [14] converted the time series of single household load data into multiple time series of residual load, which could artificially enlarge the training data. This method can solve the problem caused by the lack of historical data and improve the accuracy of residential load forecasting. In order to give full play to the advantages of CNN, Andriopoulos et al. [15] proposed a method based on statistical analysis to preprocess the time series data before solving the problem of short-term load prediction. The main feature of this method is to optimize the superparameters of a neural network by using the statistical attributes of each time series data set and transform the given data set into a form that allows maximum advantage of the CNN algorithm. In addition, the literature [16] designed a CNNGRU short-term load prediction model based on the attention mechanism, and effectively classified the household load data according to the household behavior. The accuracy of this method reaches 92.06% in small-range load prediction. Compared with the general deep learning time series prediction model, the prediction time of this method is reduced by 75%.

2.1.3. Hybrid Model

A comparison of several prediction models is shown in Table 1. Although CNNs have their unique advantages in load prediction, the efficiency of LSTMs for long-term dependencies is unquestionable. Therefore, the hybrid LSTM–CNN model becomes a reliable deep learning load prediction method worthy of further study, especially when multi-step methods are considered.

Table 1. Comparison of several forecasting models.

Model	Advantage	Shortcomings	Application Situation
Classical time series model	Simple to understand, modeling time is short	Poor performance on complex time series prediction problems	Simple time series data prediction
LSTM	Able to learn long-term dependencies	Risk of overfitting the model	Process non-linear and non-stationary time series data
CNN	Features can be learned automatically	Not good for non-linear time series data	Lack of historical data, short forecast time

Jiang et al. [17] proposed a multi-task, multi-information fusion deep learning framework (MFDL) that combines long and short-term memory (LSTM) and a convolutional neural network (CNN) to learn both near-term and long-term regular behavior, learning near-term electricity consumption behavior by LSTM and long-term regular electricity consumption behavior features by CNN. Experimental results show that this method can significantly improve the accuracy of single household short-term load prediction. Guo et al. [18] also used convolutional neural network (CNN) and long and short-term memory (LSTM) to capture temporal short-term local information and long-term corre-

lation information, respectively, and used an autoregressive (AR) model as the linear component; experimental results show that this LST Net network model can better adapt to time series and complex non-linear load data relationships. It has higher prediction and generalization performance compared with ARIMA and CNN-LSTM models. And based on an improved LSTM, Syed et al. [19] proposed a hybrid model (HSBUFC) based on bidirectional and unidirectional LSTM superposition coupled with fully connected dense layers, which amplifies the advantages of unidirectional LSTMs, bidirectional LSTMs, and RNN superposition for energy consumption prediction accuracy. The model outperforms other hybrid models such as the widely used convolutional (Conv) neural network-LSTM and ConvLSTM in terms of accuracy.

In summary, there have been many studies proving the usefulness of neural network prediction models in household load forecasting, and hybrid models based on the fusion of multiple neural networks are being developed more frequently. Future research should explore better deep learning models, such as deeper, wider, and more complex neural networks, or use new models such as self-attentive mechanisms to improve prediction accuracy and stability. Due to the volatility of loads, single forecasting models sometimes fail to achieve high accuracy. Combining forecasts using multiple deep learning methods can effectively improve model generalization. Meanwhile, in household load forecasting, multiple tasks such as power forecasting, electricity consumption forecasting, and power forecasting may be involved. Therefore, researchers can explore how to use multi-task learning methods to solve multi-task prediction to improve the prediction performance of the model while ensuring the prediction accuracy of the model.

2.2. Feature Extraction

When applying deep learning algorithms for load prediction, the input training set X-train can be fed directly into the LSTM network as if no feature extraction was performed, or it can be first transformed into a new feature space and then the transformed X-train matrix is applied to the LSTM network [20]. It is worth noting that the original load curve consists of many points, which may contain a large amount of redundant information. All lagged load values cannot be used for load prediction at subsequent points. In addition, the time series may contain useful information about the relationship between loading points that is not apparent in the original feature space [21,22]. Therefore, the application of appropriate feature extraction to the variable vector of the load curve can show the dependence, correlation, and inherent characteristics of the load curve, which is very important to improve the accuracy of load prediction.

Figure 3 shows the electricity consumption of a washing machine in a family for a week. It is not difficult to see that not only different electrical appliances have different load curves, but also the load curves of a single electrical appliance differ on different days of a week. How to extract the highly correlated features from the load curve is very important. As shown in Figure 4, the literature [3] uses the advantages of wavelet decomposition (DWT) and cooperative representation (CRT) to extract the features of hysteresis load variables. DWT can effectively separate useless details from the approximate components of variable vectors, while CRT can extract local information of load value dependence in adjacent time instances and adopt a regularization method to control major load fluctuations adaptively to avoid unwanted changes in signals. Kiprijanovska et al. [23] extracted features from the context data (weather, calendar) and the historical load of all families in the data set to integrate multiple information sources, and calculated novel domain-specific time series features so that the system could better simulate the energy consumption pattern of families. The proposed method does not depend on the current number of households in the system and has great commercial potential for predicting electrical energy consumption at the individual household level. In the literature [24], mutual information (MI) and principal component analysis (PCA) were used to select features and reduce dimensions of multi-dimensional weather influencing factors to avoid interference from unrelated factors. Finally, three-dimensional weather variables (cumulative contribution rate of 93.93%) were

obtained from the 12-D weather influence factors, which greatly improved the prediction speed of the Co-LSTM network.

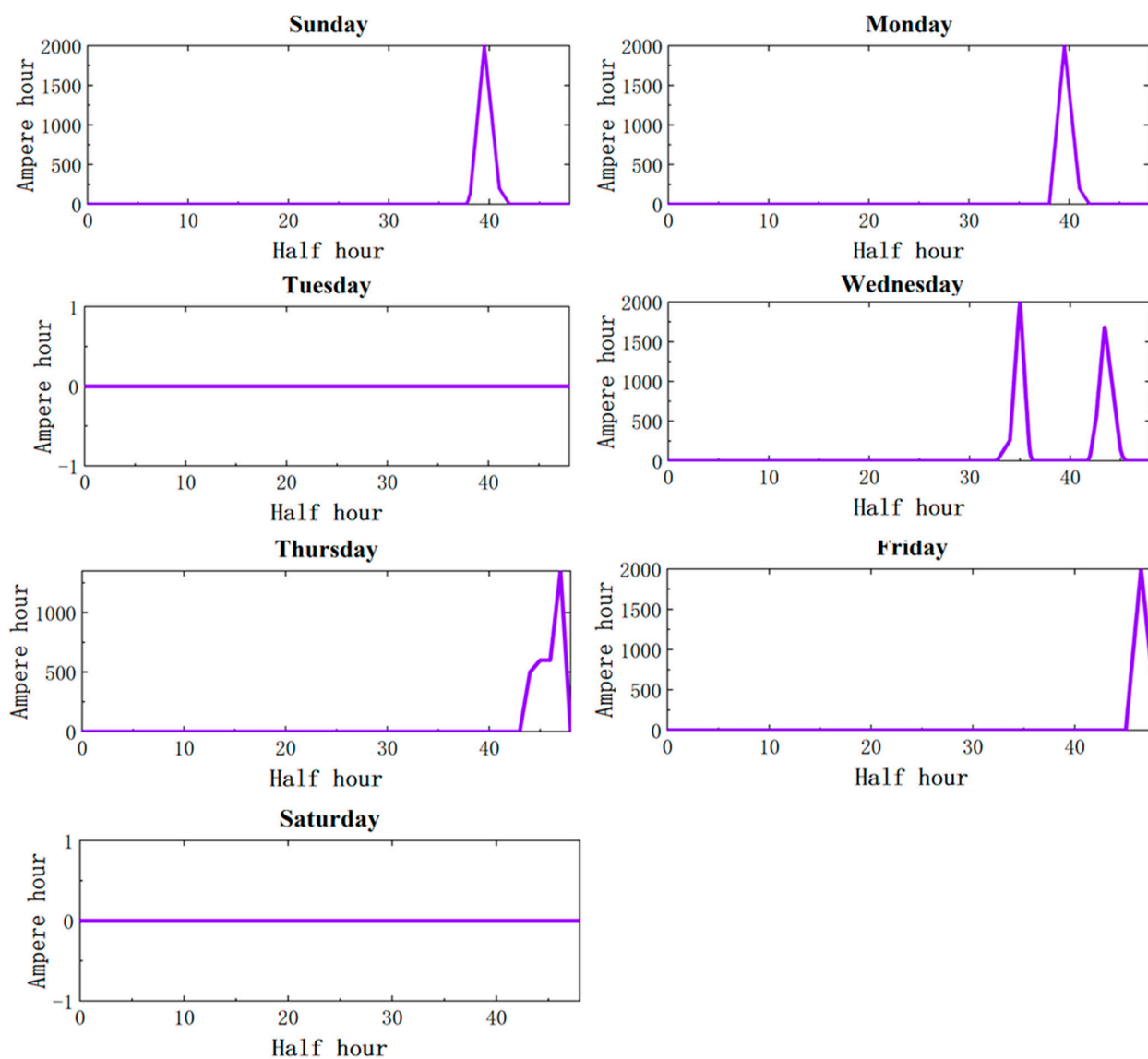


Figure 3. Consumption load of clothes washer on different days of a week.

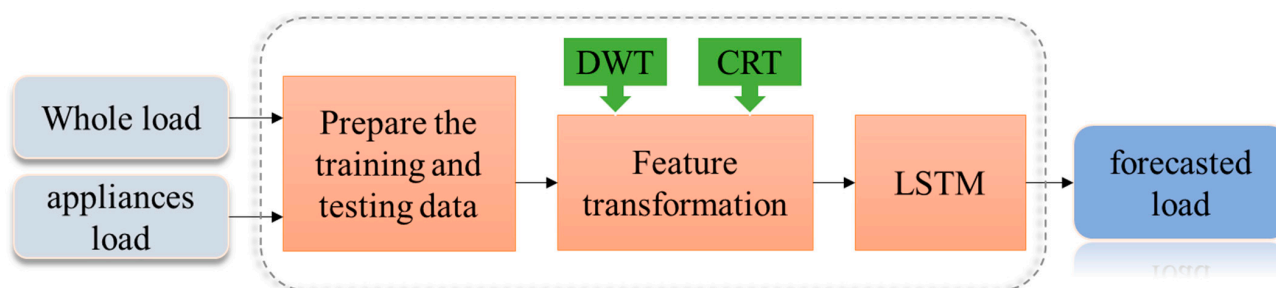


Figure 4. Proposed load forecasting framework containing preprocessing, feature extraction, and forecasting.

2.3. Adaptive Learning of Networks

The traditional static load forecasting technology obtains the single value load forecast by using the off-line training forecasting model of the consumption pattern of the historical load. However, this technology cannot capture the dynamic changes of the real load consumption pattern during the forecast period. If the real load consumption pattern changes during the forecast period, the prediction accuracy will be greatly reduced. In order to improve the shortcomings of the traditional forecasting technology, the online adaptive learning ability of the network is developed, which can effectively capture the dynamic change of load consumption, so as to improve the forecasting accuracy [25,26]. At the same time, the large-scale deployment of smart meters provides an opportunity for fine-grained load forecasting using real-time residential data.

Load forecasting for residential users is challenging because users with similar characteristics may have different consumption patterns. With the improvement of computing power, artificial neural network technology for time series prediction has become more and more widely developed. In most cases, training DNNs focuses on developing a model to predict the load consumption of a single house. To predict multiple loads on multiple houses, the standard procedure is to create a new model for each house, which results in high consumption of training time and computing power. Gomez-Rosero et al. [27] proposed a method to develop DNNs for multiple time series prediction by evolving only one architecture and adjusting the weights of each time series, a process similar to migration learning. Fujimoto et al. [28] introduced a deep ESN architecture with high affinity to edge computing, proposed an effective online learning scheme to keep prediction models up-to-date, and used real annual data from over 500 households to evaluate its effectiveness. The proposed framework will make a significant contribution in enabling predictive models otherwise suited to edge computing to exploit the benefits of deep learning architectures. Fekri et al. [29] proposed an online adaptive RNN load prediction method capable of continuously learning from newly arrived data and adapting to new patterns. The RNN is used to capture temporal dependencies, while the online aspect is continuous by adding preprocessing, buffering, and tuning modules for learning. The preprocessing module prepares the data for online learning, the tuning module adapts the neural network hyperparameters to the newly arrived patterns, and the buffering module helps to learn from particularly difficult patterns and assists with conceptual drift.

Household load forecasting is a dynamic problem and has become a topical issue to enable forecasting models to automatically update parameters according to changes in data in order to provide better accuracy and real-time load forecasting. In future work, experimenting with different adaptive algorithms (e.g., online learning, recursive learning, and incremental learning) to improve prediction models and validating their adaptive performance and prediction accuracy in real data can help us find the most suitable algorithm for the task of household load forecasting.

3. Probabilistic Load Forecasting Methods

Traditional single-value forecasts are unable to assess the uncertainty in load demand forecasts, and an accurate assessment of uncertainty is essential for applications such as optimal stochastic decision-making. Probabilistic load forecasting (PLF) methods can model the uncertainty in future electricity consumption [30,31]. Its load forecasting for individual households can be fed into robust optimization algorithms for home energy management, and probabilistic forecasting captures the probability of occurrence of each predicted value within the forecast interval, providing more comprehensive probabilistic information about future loads (Figure 5). Wang et al. [32] combined quantile regression neural networks (QRNN), quantile regression random forests (QRRF), and quantile regression gradient augmentation (QRGA) methods to select the weight-optimal process. Feng et al. [33] introduced STLF-QMS, a q-learning dynamic model selection (QMS) with a mixture of deterministic and probabilistic load prediction models. By simulating consumption scenarios, Xia et al. [34] proposed a probabilistic load forecasting method based on

consumption scenarios, where the extracted scenarios can effectively characterize household electricity consumption habits and are used to calculate the probability of occurrence of each consumption scenario for a single household in any future time horizon. In contrast, combined with deep learning, Brusaferrri et al. [35] proposed a new approach to probabilistic load prediction based on Bayesian deep learning techniques to achieve a comprehensive uncertainty characterization by capturing general patterns of stochastic uncertainty and cognitive uncertainty counterparts. A first step towards a comprehensive exploration of Bayesian hybrid density networks for probabilistic load prediction was initiated.

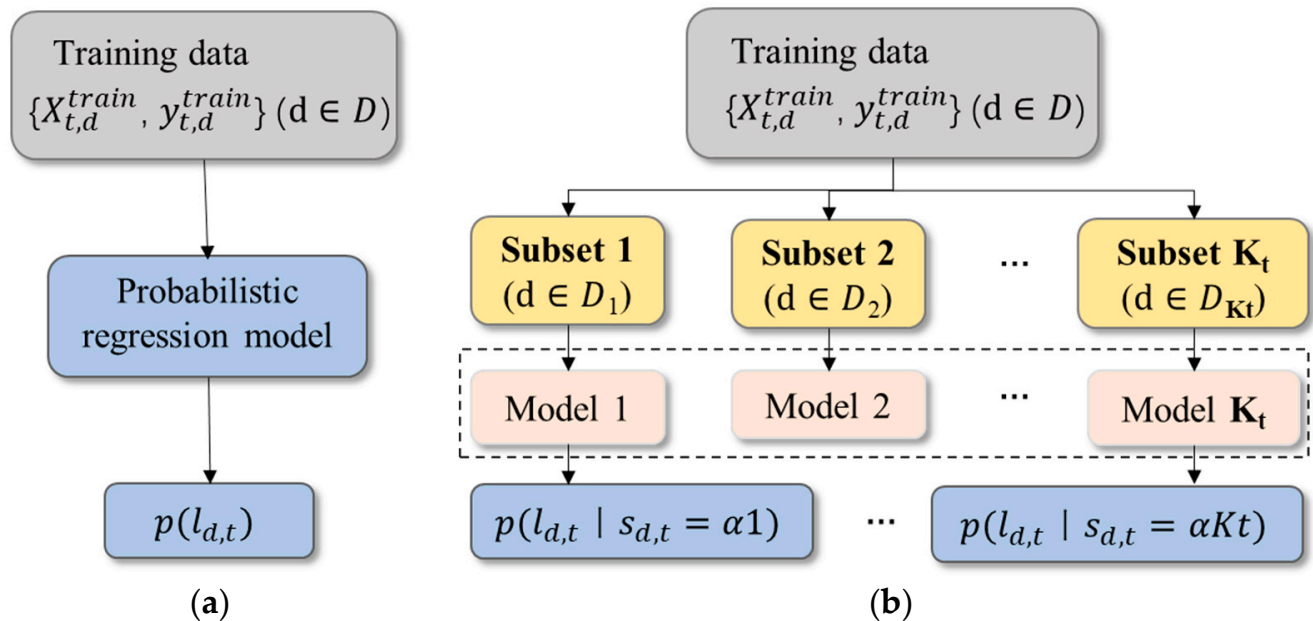


Figure 5. (a) Framework of probabilistic load forecasting. (b) Framework of conditional probabilistic load forecasting.

In conclusion, probabilistic load forecasting methods have the advantages of reliability, flexibility, scalability, and real-time in household load forecasting, but traditional probabilistic load forecasting methods based on statistical modeling are not well adapted to a wide range of complex load forecasting problems. The authors argue that combining probabilistic forecasting methods with deep learning time series models (e.g., probabilistic forecasting methods based on empirical modal decomposition), which consider both forecasting accuracy and deeply reflect electricity consumption habits, has very broad research prospects.

4. Exploration of Ultra-Short-Term and Facility-Level Load Forecasting

Much of the work in household load forecasting has previously focused on forecasting the load of the entire household. While total load forecasting is important, the distribution of household loads is often affected when customers take individual actions to save energy or shift loads, for example, by delaying the operation of dishwashers to nighttime and thus avoiding peak electricity consumption. Therefore, the development of short-term load forecasting models at the residential appliance level is also important.

4.1. Bottom-Up Approach to Household Load Forecasting

In recent years, advanced metering infrastructure (AMI) [36] has been widely deployed, while the development of artificial intelligence has greatly improved the accuracy of non-intrusive load monitoring (NILM) [37]. With the collaboration of AMI and NILM, a wealth of accurate device-level energy consumption information is available, and a bottom-up load forecasting framework based on this underlying data from domestic appliances is

a new way of thinking about forecasting. The bottom-up approach can significantly reduce forecasting errors, but also requires room and appliance-level electricity consumption data.

For individual households, energy consumption is often determined by the habits of the residents, but their behavior is accompanied by a degree of randomness due to a variety of factors. In general, however, similar factors will have a similar effect on the consumption of household appliances. Therefore, it is feasible to model household consumption behavior using statistical characteristics of appliance consumption in similar historical periods. Gao et al. [38] proposed a bottom-up forecasting framework based on the underlying data of the electricity consumption of household appliances. In the proposed framework, load distributions for group households are obtained using a similar day's extraction module (considering external environmental and internal household factors affecting energy consumption), a household behavior analysis module (developing probabilities of appliance consumption behavior), and a household behavior prediction module.

However, unfortunately, this framework has only been verified in the group data of multiple groups of families, which has a better prediction effect on the load distribution of group families. For a single family, Zheng et al. [39] et al. verified that the bottom-up method equipped with the long- and short-term memory cyclic neural network (LSTM) model could obtain better results than the direct prediction at the family level. However, for practical applications of the bottom-up approach, the LSTM model is inefficient and expensive. In order to effectively model a variety of different device usage patterns, a more realistic lightweight algorithm is applied—the Kalman filter model adaptively predicts device usage with less historical data and shorter computational time, outperforming large algorithms such as the LSTM model. The literature [40] proposed a bottom-up method considering the load characteristics of the equipment level. Considering the working principle and load characteristics of home appliances, home appliances were divided into continuous load and intermittent load. Through the accumulation of home appliance loads, more refined load prediction could be achieved. At the same time, considering the fluctuation of load curve, the loess-based seasonal trend decomposition program (STL) was applied to continuous load appliances to obtain finer grain and more regular load data. The model can predict multiple equipment simultaneously and improve the computational efficiency.

In the HEMS, predictions of short-term future electricity consumption are used to support HEMS decision-making. Although there have been many studies on the prediction of building electricity consumption, there is a lack of spatial analysis of predictive performance, especially at the household level, the appliance level, or the submeter level. Zheng et al. [4] adopted a LSTM neural network to establish a multi-step load prediction model, as shown in Figure 6. The influence of spatial factors on the forecasting accuracy of household electricity consumption was analyzed. The results showed that the prediction accuracy can be improved by predicting the electricity consumption of household appliances first and then by aggregating, compared with predicting the electricity consumption of households directly. Using deep learning technology to train the network based on historical data of energy use of residential appliances, it is also possible to estimate the energy use of a given appliance in the short term. Razghandi et al. [41] proposed a residential electricity load forecasting model. This model used an LSTM cyclic neural network to learn the energy consumption pattern of individual appliances in smart homes. After preprocessing the data collected by smart meters, we inputted it into the deep neural network to generate the predicted load consumption of devices and predicted the potential load consumption of a given appliance at different time intervals.

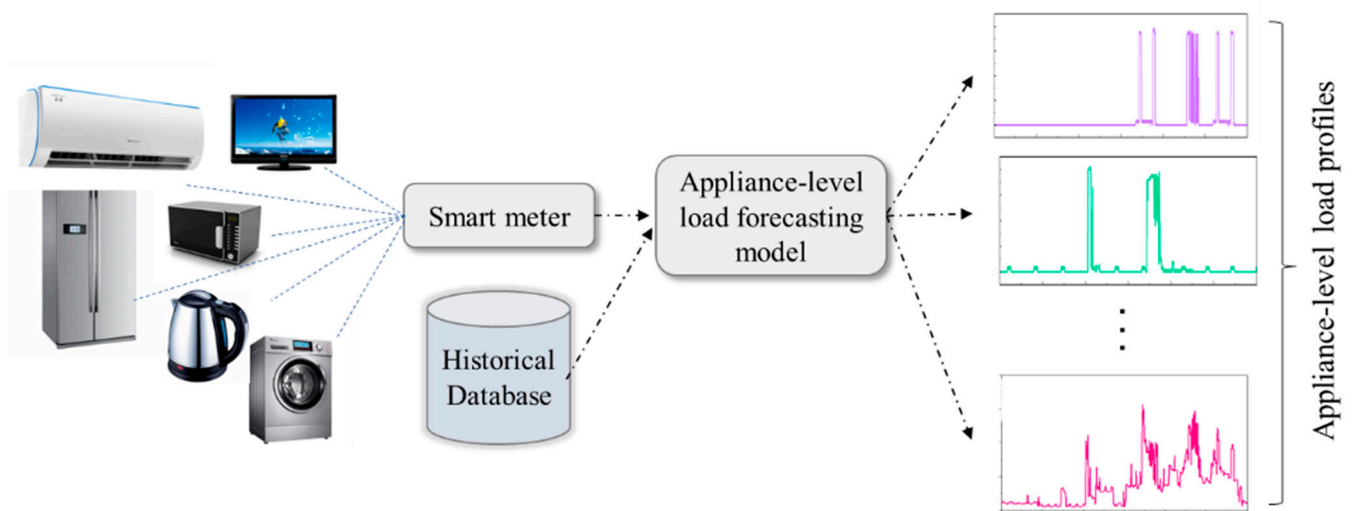


Figure 6. Framework for the proposed appliance-level load forecasting deep learning-based model.

A bottom-up load prediction method requires accurate load prediction for each household appliance. This kind of device-level load prediction is a very challenging task. Previous studies have provided several forecasting frameworks, although they all have had good simulation effects [42,43]. However, their feasibility still needs to be studied. The deep learning algorithms used by the framework are relatively simple, and most of them cannot give consideration to both accuracy and efficiency. According to the author, from the power performance of equipment, the living habits of household users, the structural characteristics of intelligent prediction algorithm, and so on, the probabilistic prediction method considering load uncertainty is integrated, and the improved LSTM neural network which has proved to be the most advanced load prediction algorithm is used. It is a worthy research direction to develop a set of equipment-level load forecasting models which take into account both prediction accuracy and practical feasibility.

4.2. The Challenge of Hourly Load Forecasting

On the other hand, with the deployment of IOT devices in buildings, electricity load usage can be collected in real time, and the real-time data collection and processing makes ultra-short-term load prediction at the hour level possible [44]. The literature [45] uses BN to simulate the relationship between variables and proposes a multi-family ultra-short-term load forecasting method that does not use a single generalist model and multivariate data for load aggregation. Ageng et al. [46] proposed an hourly load prediction framework combining data preparation and LSTM. Firstly, interpolation and Savitzky–Golay filtering were carried out on the collected data so as to extract the appropriate load consumption mode for the next hour's load prediction. In addition, in order to better determine the relationship between the time series information, a two-layer LSTM was used.

It has been found from a large number of studies that in the ultra-short-term household appliance load prediction, data-driven models such as artificial neural networks can predict the trend of electric energy consumption of electrical equipment but often ignore the consumption peak [47], that is, the switch state prediction of electrical equipment is more accurate, while the peak prediction still has defects, as shown in Figure 7.

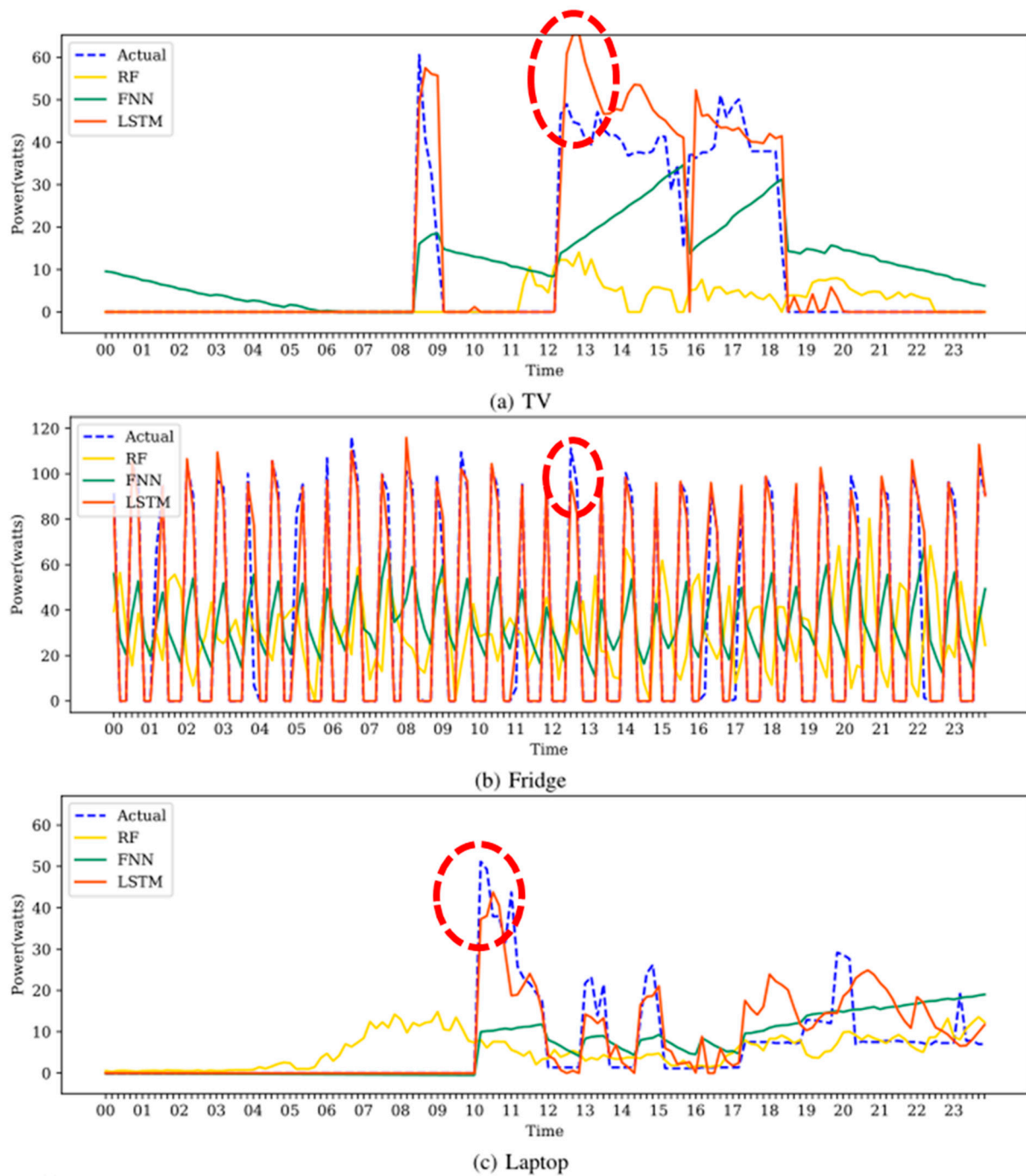


Figure 7. Load prediction results of different electrical equipment based on data-driven model.

It is helpful to predict the load consumption in the next hour of the household to correctly manage household energy [48]. However, ultra-short-term load prediction is not easy to achieve, mainly for the following reasons: (1) The dynamic behavior of each user will affect the load usage pattern, which plays a major role in the hourly load prediction. (2) Transmission and technical errors will also affect load usage patterns, thus reducing the accuracy of ultra-short-term load prediction. (3) Inevitably, data loss or overflow may occur due to hardware or system errors. Noise can also be added to the collected data. In this case, the accuracy of the load forecast for the next hour will be reduced.

In future research, the method of hyperparameter adjustment can be developed to improve the prediction accuracy. Secondary influences on load consumption, including the number of people, season, and holiday period, can also be further studied.

5. Load Forecasting Is Used for Electricity Scheduling in the HEMS

Recently, the home energy management system (HEMS) as the main energy saving countermeasure has gradually become the focus of residential energy optimization research. In the context of the smart grid, the HEMS, as a hub connecting users with the power grid, aims to optimize the utilization of electric energy by household users [49,50]. Table 2 shows part of the data collected by smart meters. The main function of the HEMS is to optimize household energy use, improve energy efficiency and user comfort through optimization algorithms and scheduling strategies, and achieve energy saving and consumption reduction by relying on the scheduling of distributed energy and energy storage devices through detection facilities such as smart meters and sensors [51]. The HEMS structure is shown in Figure 8. According to the information of users' electricity demand, environmental conditions, and price incentives, the built-in residential electricity optimization strategy is applied to adjust the operation of various electrical appliances, optimize the user load curve, and participate in peak regulation. Household load forecasting can predict household electricity demand in advance, help households better plan energy use, and reduce electricity costs and unnecessary energy waste. Schmidt et al. [4] proposed a probabilistic model for predicting the electrical consumption of household appliances based on historical data, taking into account three main factors: (1) the time elapsed after the last use of household appliances; (2) the time of day during which the appliance may be used; (3) behavioral patterns of occupants (for example, people are likely to use the washing machine two to three times more on weekends than weekdays). The authors aim to provide advice to occupants and improve home automation systems for load shifting or scheduling without affecting the comfort of occupants.

Table 2. A segment of the data collected by smart meters.

Time	Active Power (kW)	Reactive Power (kW)	Average Voltage (V)	Average Current Intensity (A)	Total Load (W·h)
16 December 2016 17:24:00	4.216	0.418	234.840	18.400	18
16 December 2016 17:25:00	5.360	0.436	233.630	23.000	17
16 December 2016 17:26:00	5.374	0.498	233.290	23.000	19
16 December 2016 17:27:00	5.388	0.502	233.740	23.000	18

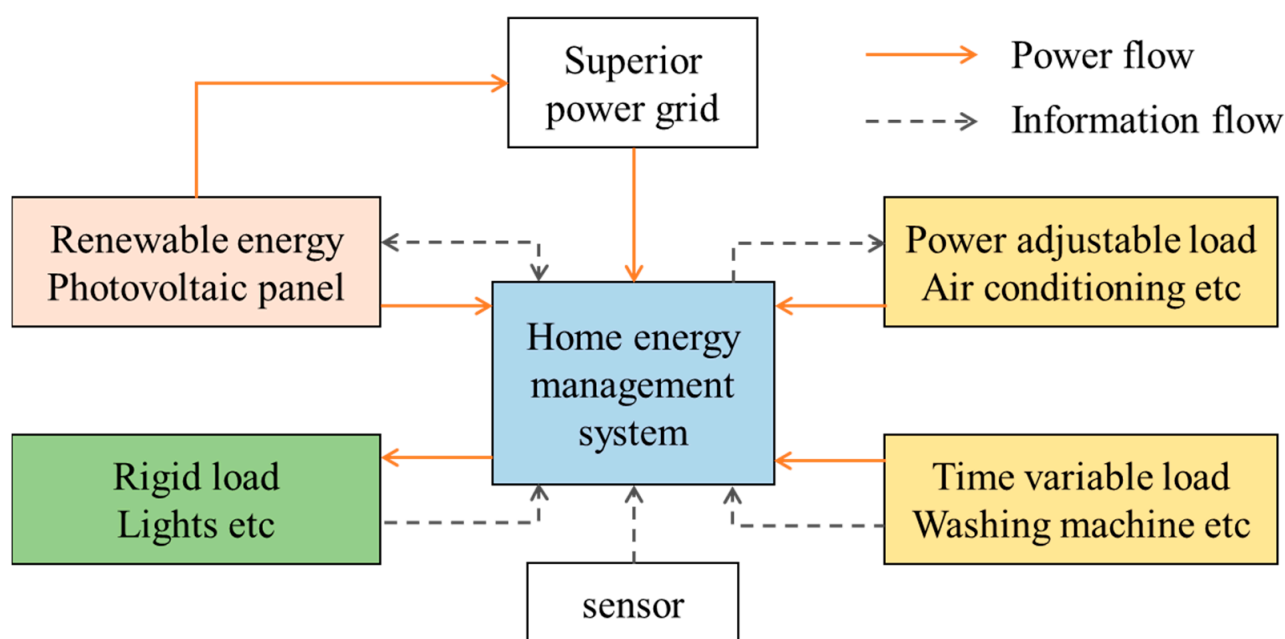


Figure 8. Schematic diagram of the home energy management system.

The HEMS can dispatch energy consumption in households through real-time electricity prices. It should be noted that although power companies release price information to customers through smart meters or the network [52], the price data are usually released several hours in advance for the next time period. Therefore, when scheduling using the HEMS, the electricity price data for the scheduling period may be partially unknown. In this case, only with the short-term forecasting function, the HEMS can better adjust the working state of power equipment according to the energy consumption data and future electricity price, so as to meet the needs of users and reduce the system operation cost. Load prediction error will increase the uncertainty of the HEMS and affect the dispatching performance of the household energy management system [53–58]. Therefore, a mature household energy management system must be equipped with a set of efficient household load forecasting models.

According to the different consumption modes and user demand, household electricity load can be divided into dispatchable load and undispachable load. These include high-voltage AC transmission systems, washing machines, clothes dryers, water heaters, dishwashers, electric vehicles, pumps for home swimming pools, etc. Under certain conditions, when the operation of this equipment scheduling will not affect the comfort of users, they are schedulable loads. The unschedulable loads mainly include computers, refrigerators, lighting systems, security systems, etc. The scheduling of these loads will seriously affect the satisfaction of users. Therefore, optimizing the scheduling of the schedulable load is one of the important ways to realize the demand response of residents. However, predicting the power consumption of a schedulable load such as a washing machine is a challenging task.

In the future, with the popularity of the smart home and the development of the HEMS, household appliance load prediction will play a more important role. Future research directions include improving prediction accuracy, improving real-time performance and stability of algorithms, and developing more intelligent and sustainable energy management systems.

6. Summary and Prospect

At present, the energy consumption of the household as a unit is increasing. In order to make rational use of energy and optimize user cost, energy management with the HEMS as the main energy-saving countermeasure has gradually become the focus of residential energy optimization research. Accurate short-term household load prediction can effectively improve the energy management efficiency and reliability of the HEMS, so as to achieve the goal of efficient utilization of energy and energy saving and emission reduction. In this paper, we reviewed the latest research results of the deep neural network load prediction method and probabilistic load prediction method in household load prediction, and summarized and analyzed the network prediction model, feature extraction method, adaptive learning ability of the network, equipment-level load prediction method, and other aspects. The key role of short-term household appliance load forecasting in HEMS dispatching was also pointed out.

Although many models and methods have achieved certain results in short-term household load prediction, there are still many problems to be solved in future research:

1. Data quality: Short-term household load forecasting requires a large amount of historical load data and other relevant data, such as temperature and humidity. However, the existing data often have quality problems, such as missing values, noise, etc., which will affect the prediction results.
2. Prediction accuracy of the model: Although the deep learning model has achieved a certain accuracy in household load prediction, there is still a lot of room for improvement. Researchers need to explore new model structures or optimization algorithms to improve the prediction accuracy.

Generalization ability of the model: The deep learning model is prone to overfitting, especially when the data set is small. Therefore, how to enhance the generalization ability of the model is an important research direction.

3. Real-time performance and uncertainty: Household load forecasting involves many uncertain factors, such as weather and user behavior. Therefore, how to consider these uncertainties and integrate them into the deep learning model is also a problem that needs to be studied.
4. Model interpretability: Deep learning models are generally regarded as black box models. In practical applications, the interpretability of models can facilitate users to better understand the prediction results of models and make corresponding decisions. However, in past studies, the explanatory forecasting framework of short-term household load is rarely involved.

To solve these problems, in the future, the research direction of short-term household appliance load prediction will include the following aspects:

1. Explore more data sources: The continuous progress of data mining and preprocessing technology can improve the quality of data and reduce the workload of data processing. The construction of data sharing and open platform will help promote the research progress of short-term household load forecasting and promote its application in practical production. In addition to electricity consumption data, other data can be used, such as meteorological data, calendar data, social media data, etc., to help forecast household load. Therefore, it is possible to consider how to combine different data sources to improve the accuracy of the prediction.
2. Model innovation: Model fusion and integrated learning technology can combine the advantages of various models to improve the prediction accuracy and robustness, such as the hybrid model of different deep learning neural networks, the model combining the probabilistic prediction method with deep learning time series model, etc. The development of intelligent technologies, such as deep learning and natural language processing (NLP), is expected to bring new breakthroughs for short-term household load forecasting and improve the forecasting effect. In addition, exploring the use of multi-task learning methods to realize synchronous prediction of multiple tasks such as power consumption and power can also improve the prediction performance of the model.

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